Università degli Studi di Padova

Dipartimento Matematica
Dipartimento di Matematica "Tullio Levi-Civita"

Business, Economic and Financial Data Final Project

A Time Series Analysis of Cinema Admissions in Italy

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1 Introduction

Over the past few years, we have grown accustomed to watching movies on OTT (Over-The-Top) streaming—services that deliver content directly to viewers over the internet. These platforms have become an integral part of our daily routines, allowing us to enjoy movies in the comfort of our homes. However, this shift has made us forget the unique joy of sharing emotions with a crowd in a movie theater.

Particularly after the COVID-19 pandemic, it seems like the streaming platforms became dominant, and the overwhelming volume of content on these services has made it increasingly difficult for individual films to draw attention in cinemas. This led me to wonder: is traditional cinema culture approaching its end?

To explore further, this study focuses on the time series analysis and forecasting of cinema admissions in Italy.

1.1 Problem Definition

This study aims to address the following questions:

- Is traditional cinema culture coming to an end?
- Will OTT services dominate the distribution of movies in the future?
- Have the COVID-19 pandemic and/or OTT platforms impacted theater admissions significantly?

1.2 The Data Sources

The dataset used in this study consists of monthly data spanning from January 2006 to December 2023, complemented by a simplified annual dataset that includes only OTT Subscriptions and OTT Spending.

The data sources are as follows:

- **Società Italiana degli Autori ed Editori (SIAE):**
  Number of screenings in theaters, cinema admissions, ticket prices, audience spending [1].
- **International Video Federation (IVF) & Statista:**
  OTT subscriptions and OTT spending (annual data only) [4, 5].
- **Istituto Nazionale di Statistica (ISTAT):**
  Consumer Price Index (CPI) (Month-over-Month) for all items and CPI (MoM) for recreation and culture [3].
- **Istituto Superiore di Sanità (ISS):**
  Number of reported COVID-19 cases [2].
- **Google Trends:**
  Search interest for OTT platforms, used as a proxy for streaming consumption behavior [7].

2 Data Exploration & Preprocessing

2.1 Dataset Overview

This study uses two datasets: a monthly dataset (January 2006 - December 2023) and an annual dataset (2006 - 2023). These datasets provide detailed insights into cinema admissions, audience spending, economic indicators, and trends related to online streaming services (Netflix, Amazon Prime Video, etc.).

2.1.1 Monthly Dataset

The monthly dataset serves as the primary source for time series modeling. The key variables included are:

- **Cinema Admissions:** The number of individuals attending cinemas each month.
- **Number of Screenings:** The total number of movie screenings conducted per month.
- **Cinema Audience Spending:** The total expenditure by cinema-goers, including ticket sales and other expenses (in euros).
- **Average Spending Per Person:** The average amount spent per visitor in a given month (in euros).
- **Google Trends OTT:** A measure of online search interest in OTT (Over-the-Top) platforms. According to Statista's study on video-on-demand platforms, the most popular SVOD services in Italy as of June 2023 was Netflix, second came Amazon Prime Video, and Disney+ followed behind. [6] Therefore Google Trends for these platforms serve as an estimate of consumer engagement with streaming services.
- **CPI (All Items):** The month-over-month Consumer Price Index for all goods and services, reflecting general inflation.
- **CPI (Culture):** The CPI specific to cultural and entertainment-related expenses.
- **COVID-19 Cases:** The number of reported COVID-19 cases per month.

2.1.2 Annual Dataset

The annual dataset provides OTT Subscriptions, OTT Spending in annual format and aggregated yearly data on cinema attendance. This dataset is used to examine long-term competition in cinema admissions and OTT subscriptions. The key variables included in the annual dataset are:

- **OTT Subscriptions:** The total number of active OTT platform subscriptions per year.
- **OTT Spending:** The total consumer expenditure on OTT services (in euros).
- **Cinema Admissions:** The yearly total of cinema admissions.

2.2 Handling Missing Values

The dataset initially lacked monthly data for cinema-related features (audience admissions, number of screenings, audience spending, average spending per person) for the year 2013, with only the total annual values available. To address this issue, monthly values were imputed using a proportion-based approach to ensure consistency with historical trends and seasonality. To impute the missing values, monthly averages from the adjacent years (2011, 2012, 2014, and 2015) were used to calculate proportional distributions. These proportions were then applied to the total annual values for 2013.

Additionally, a small number of missing values in 2020 and 2021 (likely due to disrupted operations during the COVID-19 period lock-downs) were replaced with the average ticket price for those respective years.

The annual dataset doesn't have missing values.

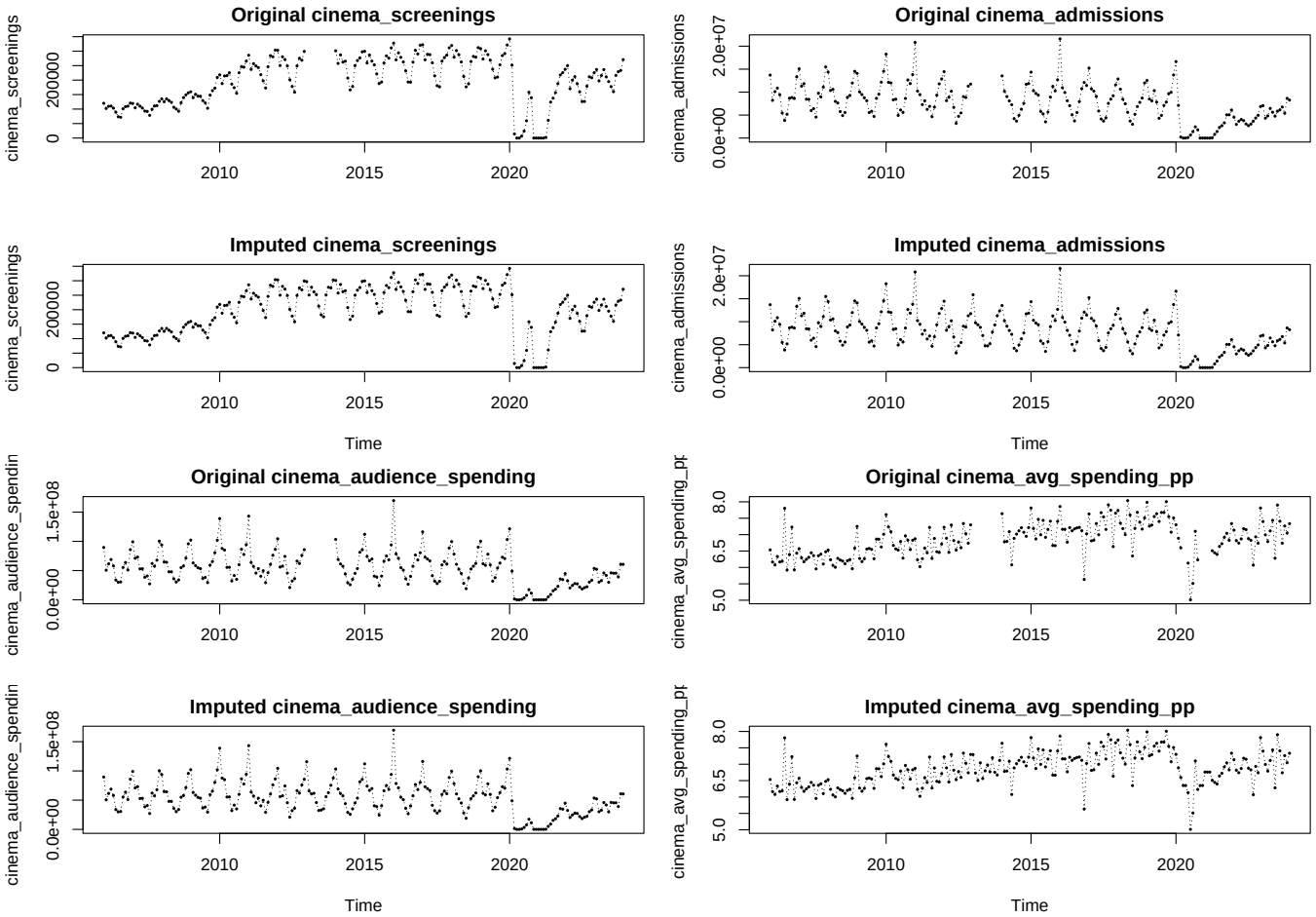


Figure 1: Original vs. Imputed Dataset

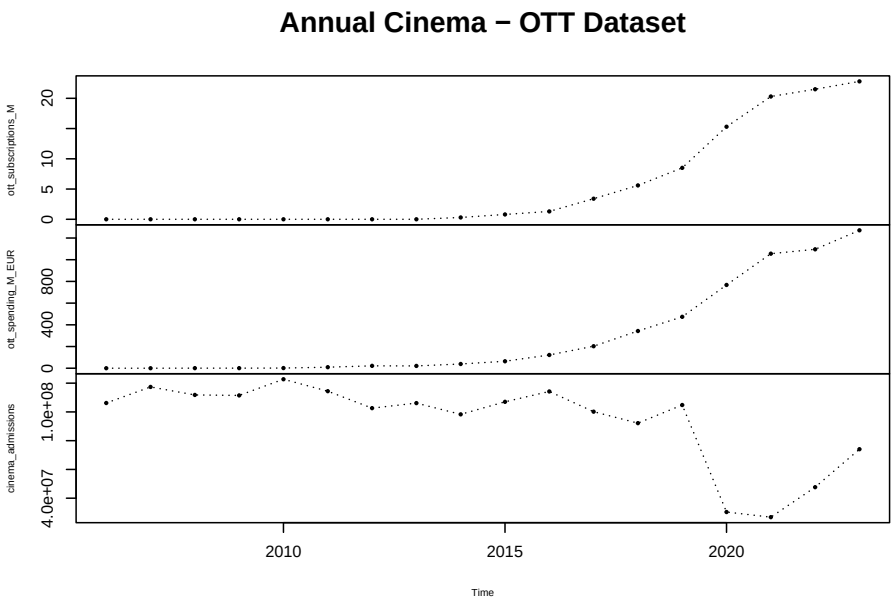
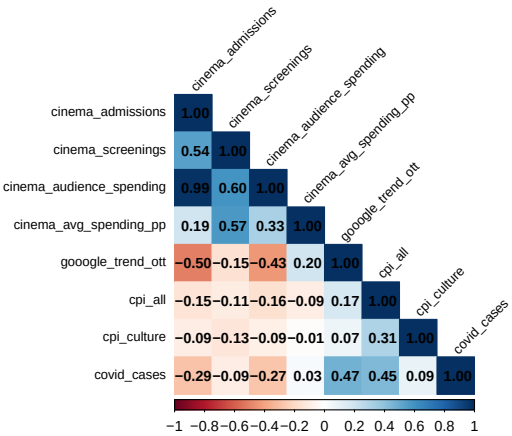


Figure 2: Overview of the Annual Dataset

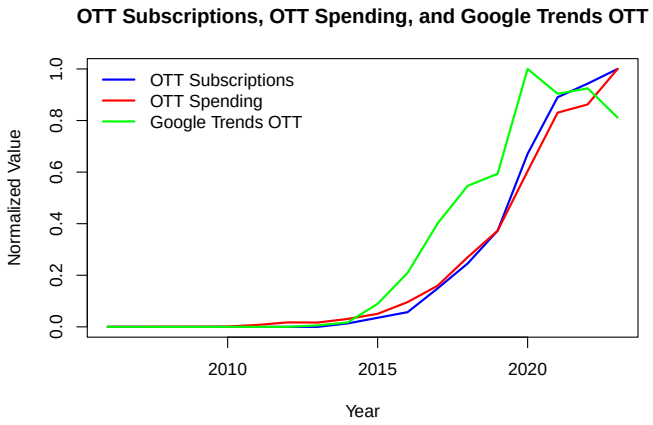
2.3 Collinearity Check

2.3.1 Correlation Matrix for Monthly Data

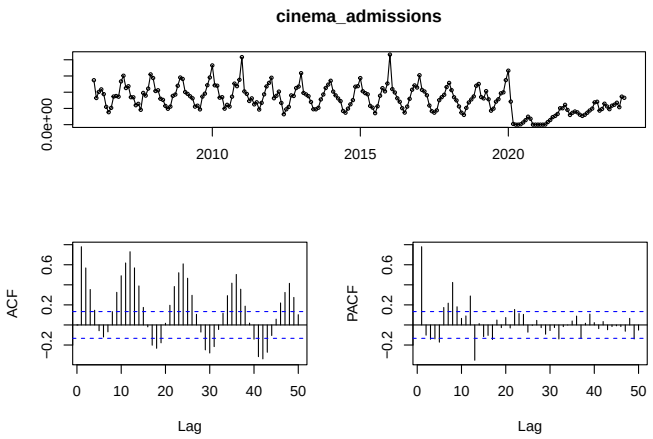


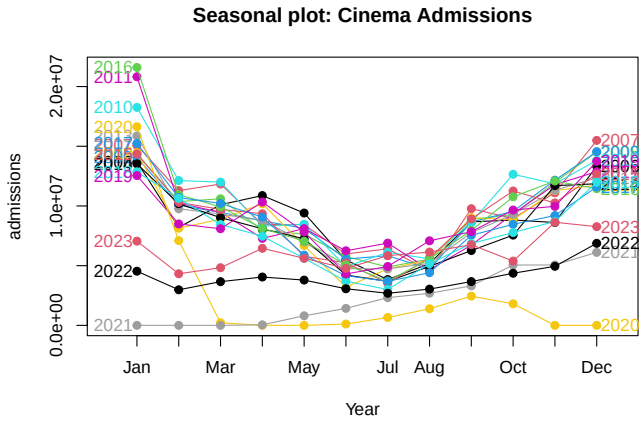
2.3.2 VIF Analysis for Monthly Data

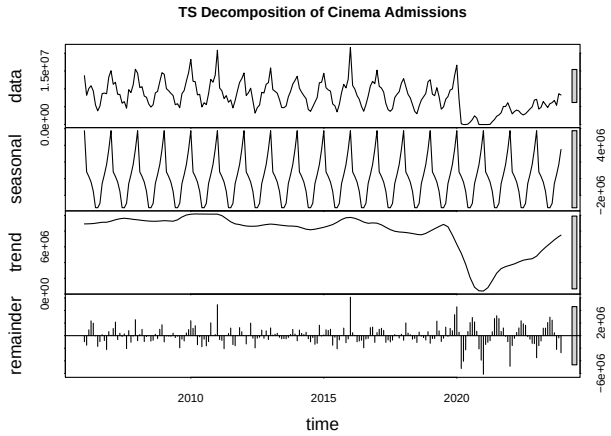
2.3.3 Check correlation & trends between OTT Data (Annual) and Google Trends OTT (Monthly)



2.4 Time Series Decomposition



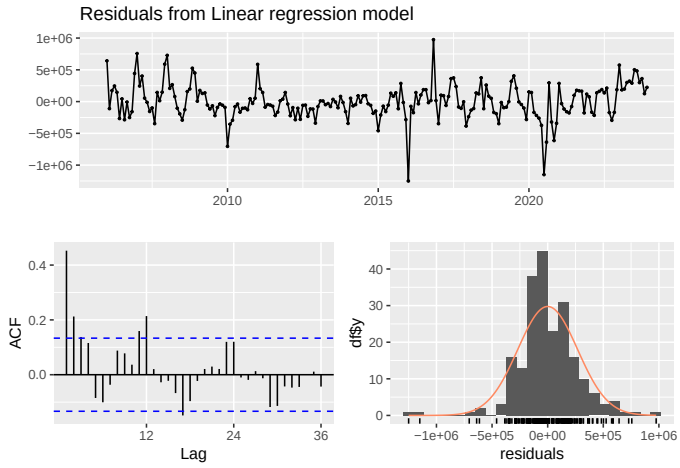




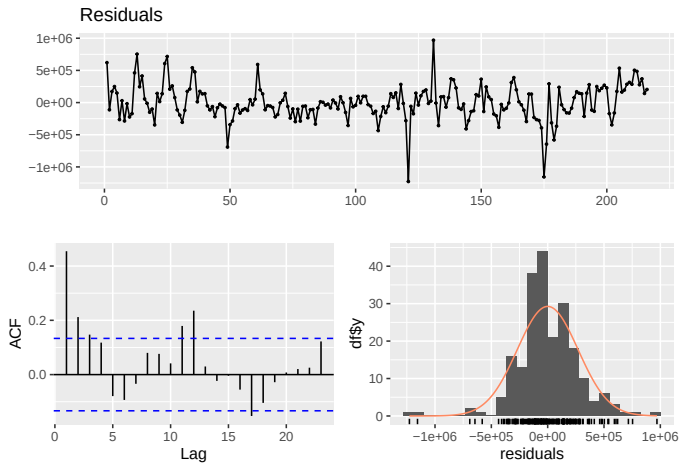
3 2) Time Series Forecasting

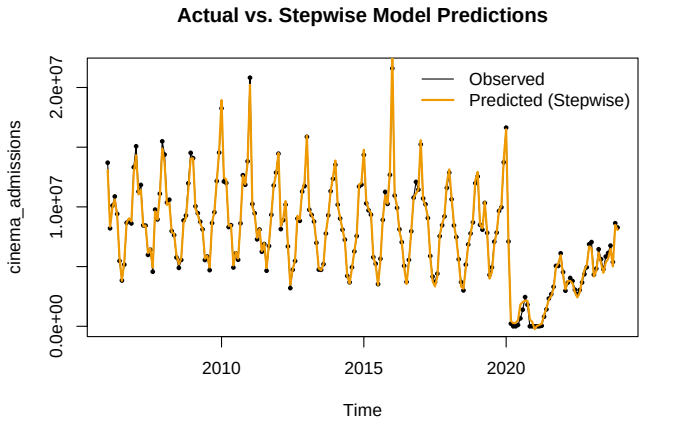
3.1 Linear Regression

3.1.1 Multiple Linear Regression with Time Series



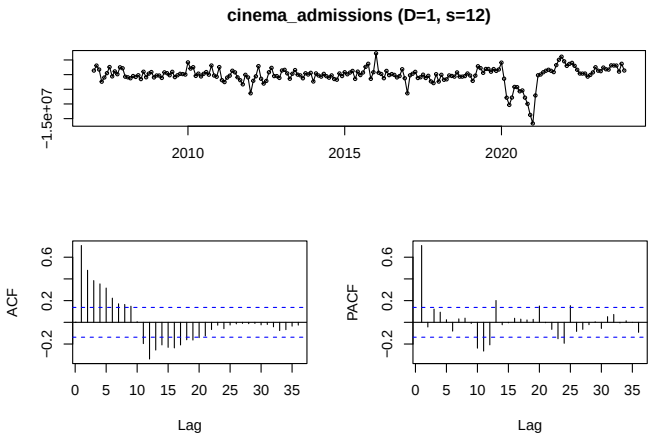
3.1.2 Stepwise Regression

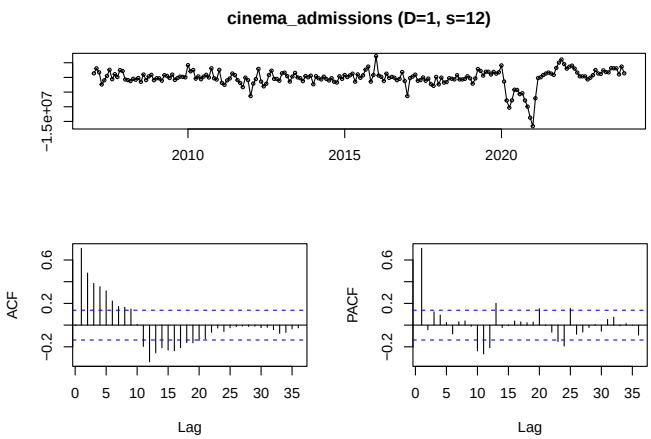


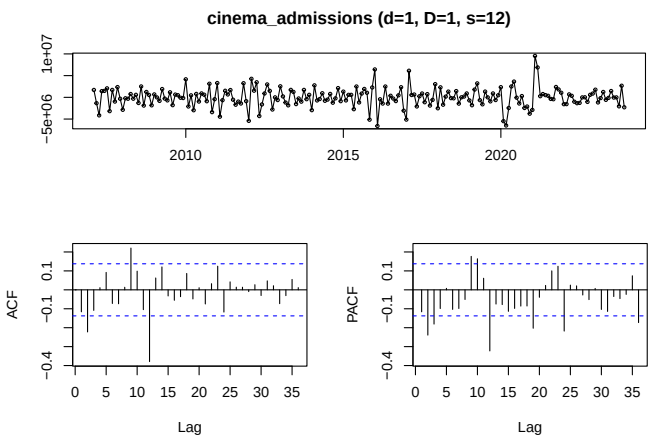


3.2 ARIMA Models

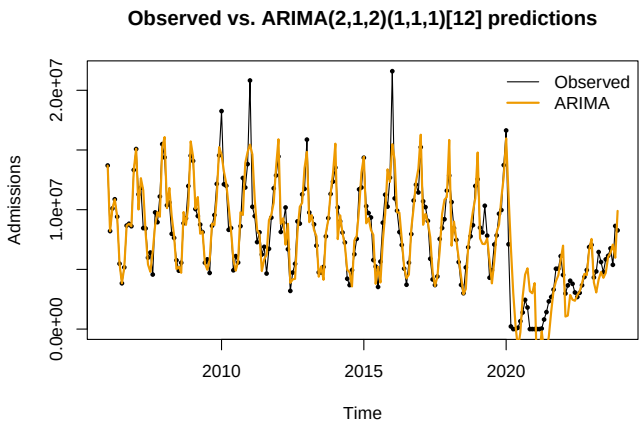
General indication (for differenced data): - If the ACF is exponentially decaying or sinusoidal and there is a significant spike at lag p in PACF and nothing else, it may be an ARIMA(p,d,0). - If the PACF is exponentially decaying or sinusoidal and there is a significant spike at lag q in ACF and nothing else, it may be an ARIMA(0,d,q).

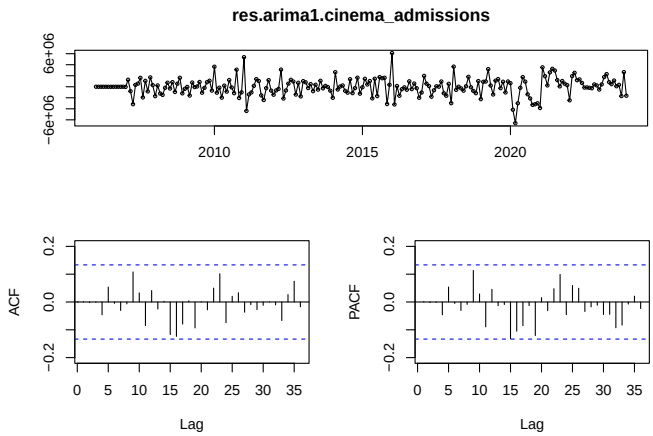


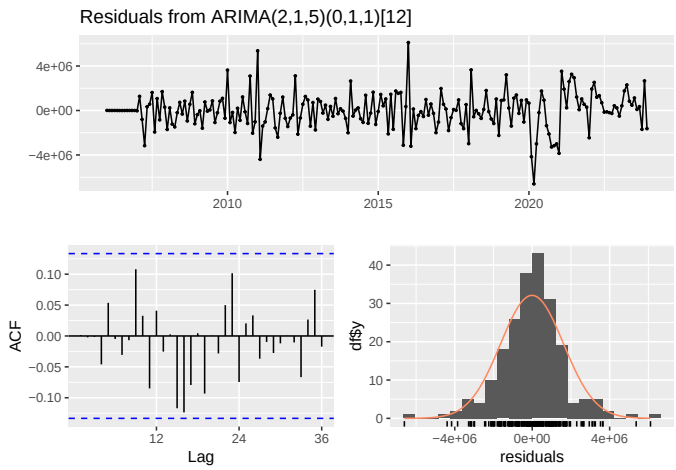


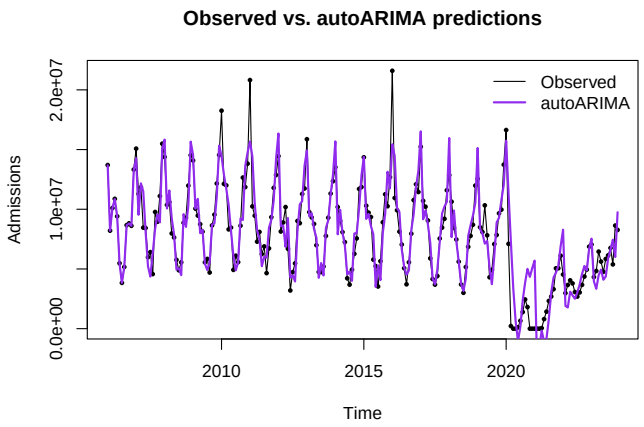


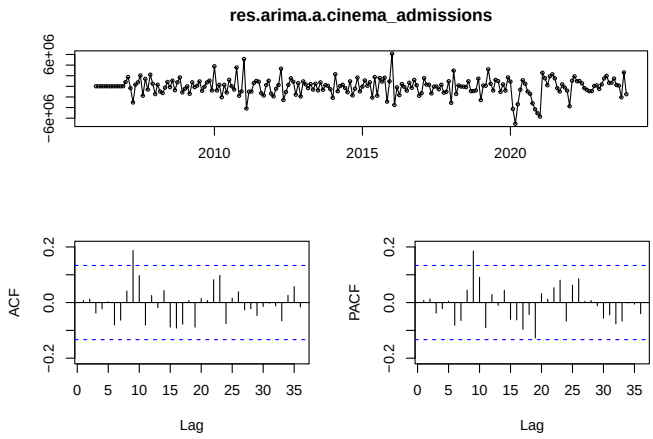
3.2.1 ARIMA


Observed vs. ARIMA(2,1,2)(1,1,1)[12] predictions
Observed
ARIMA
Admissions
Time


res.arima1.cinema_admissions
ACF
PACF
Lag


Residuals from ARIMA(2,1,5)(0,1,1)[12]
ACF
Lag
df$y
residuals


Observed vs. autoARIMA predictions
Observed
autoARIMA
Admissions
Time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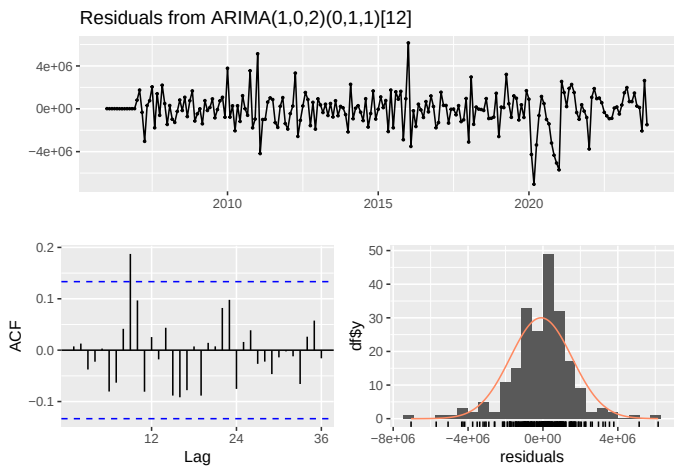
ACF
PACF
Lag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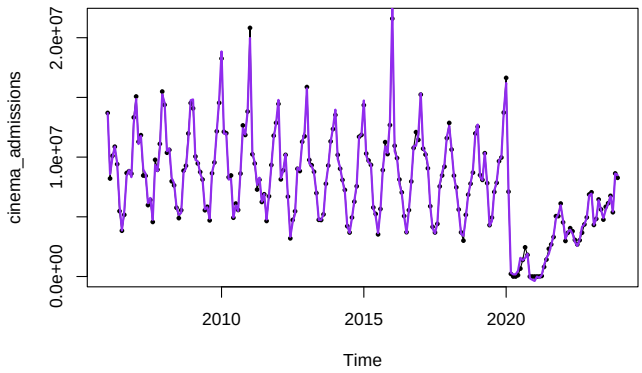
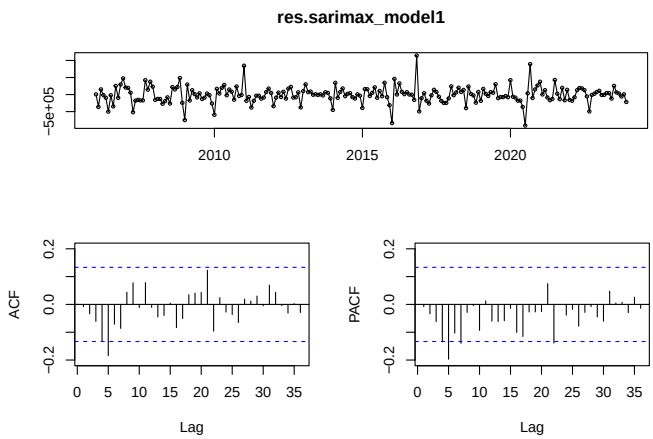
4e+06
0e+00
−4e+06
2010
2015
2020
ACF
0.2
0.1
0.0
−0.1
12
24
36
Lag
df$y
50
40
30
20
10
0
−8e+06
−4e+06
0e+00
4e+06
residuals

3.2.2 ARIMAx


cinema_admissions
2.0e+07
1.0e+07
0.0e+00
2010
2015
2020
Time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2010
2015
2020
ACF
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0
5
10
15
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Lag
PACF
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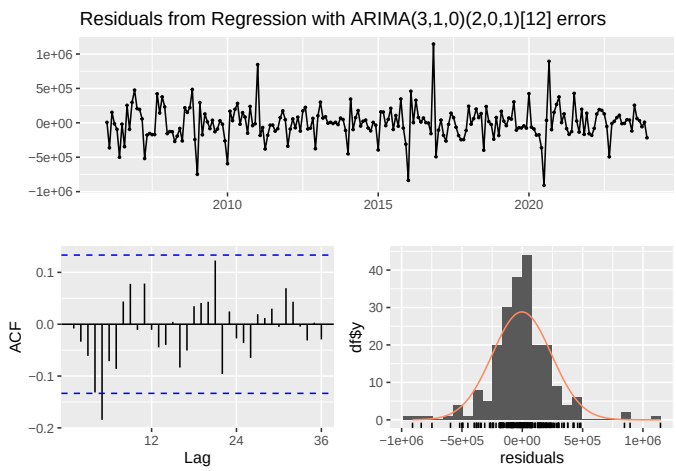
Lag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0e+00
−5e+05
−1e+06
2010
2015
2020
ACF
0.1
0.0
−0.1
−0.2
12
24
36
Lag
df$y
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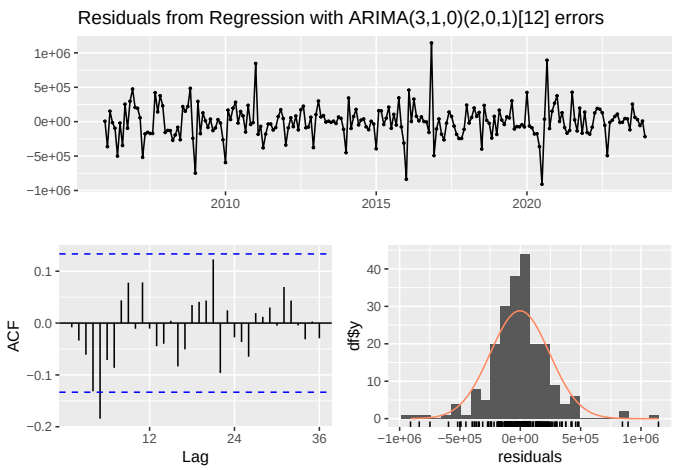
30
20
10
0
−1e+06
−5e+05
0e+00
5e+05
1e+06
residuals


Residuals from Regression with ARIMA(3,1,0)(2,0,1)[12] errors
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5e+05
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2010
2015
2020
ACF
0.1
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−0.2
12
24
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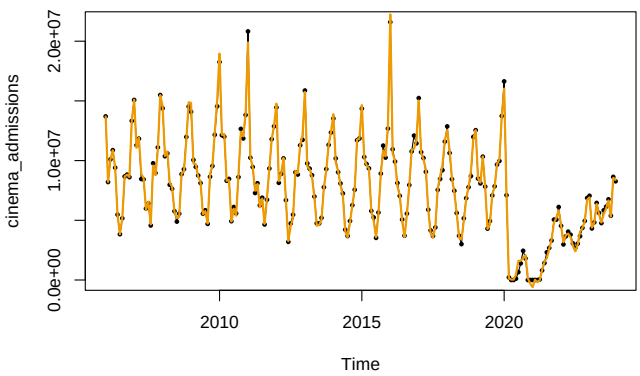
0e+00
5e+05
1e+06
residuals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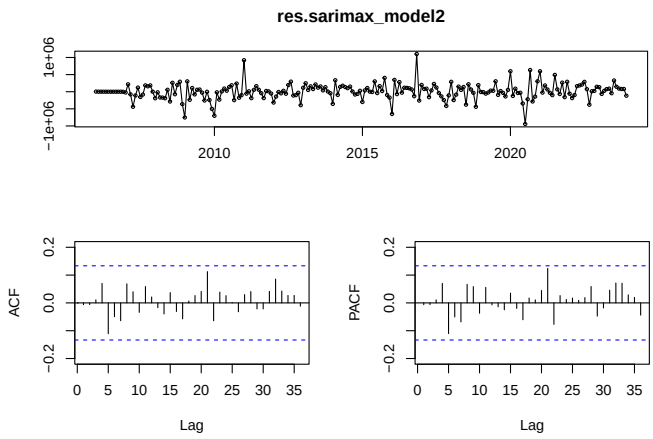
2.0e+07
1.0e+07
0.0e+00
2010
2015
2020
Time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2010
2015
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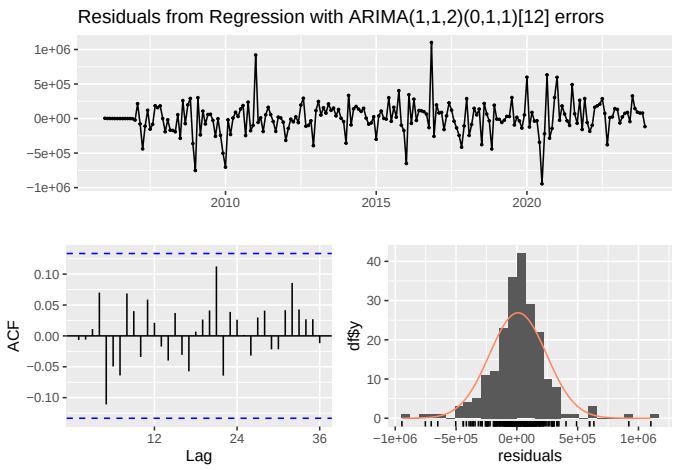
Lag


Residuals from Regression with ARIMA(1,1,2)(0,1,1)[12] errors
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0e+00
−5e+05
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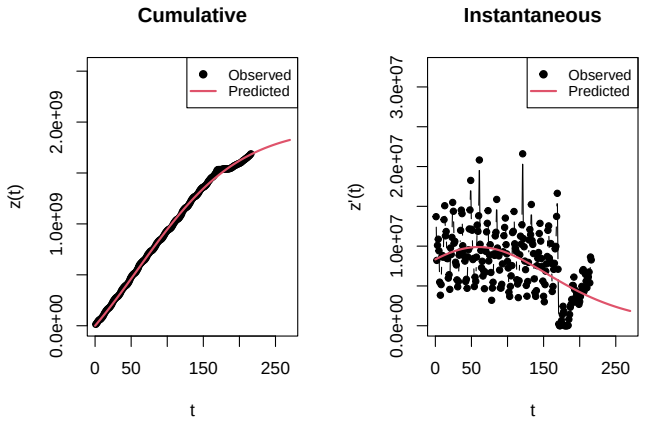
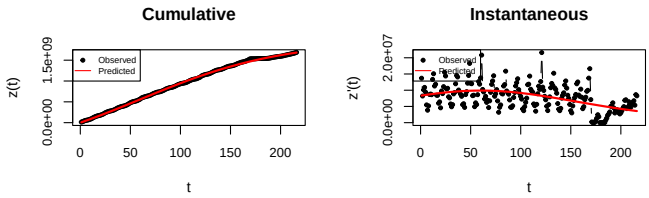
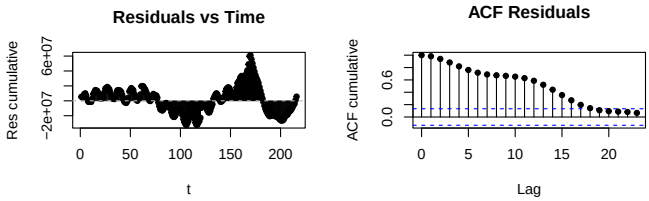
2010
2015
2020
ACF
0.10
0.05
0.00
−0.05
−0.10
12
24
36
Lag
df$y
40
30
20
10
0
−1e+06
−5e+05
0e+00
5e+05
1e+06
residuals

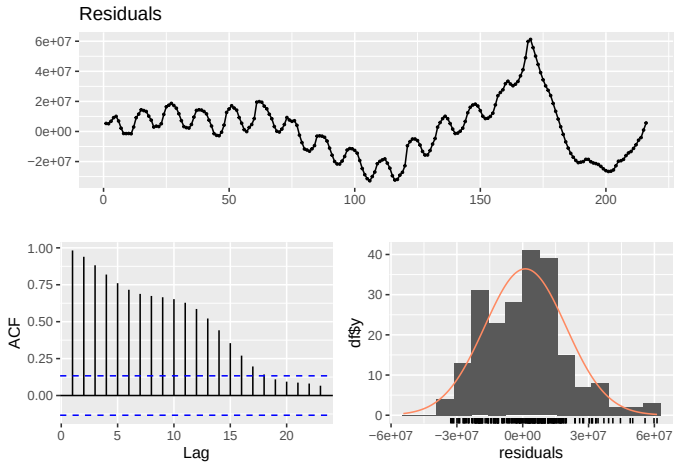
3.3 Diffusion Models

3.3.1 Bass Model (BM)



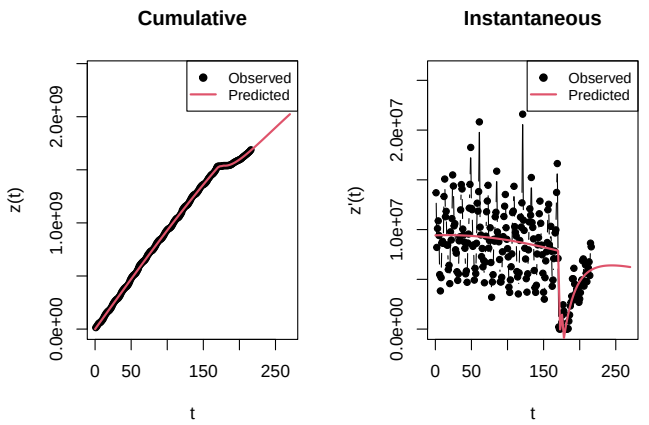


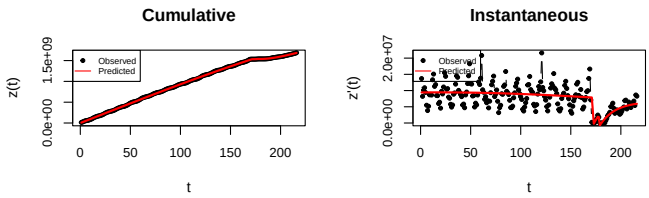


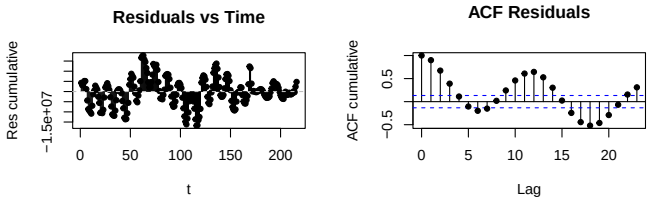


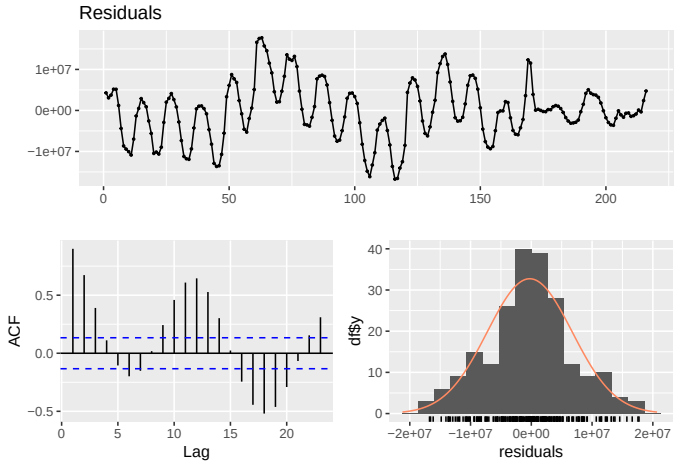
Future Prediction

3.3.2 GBM

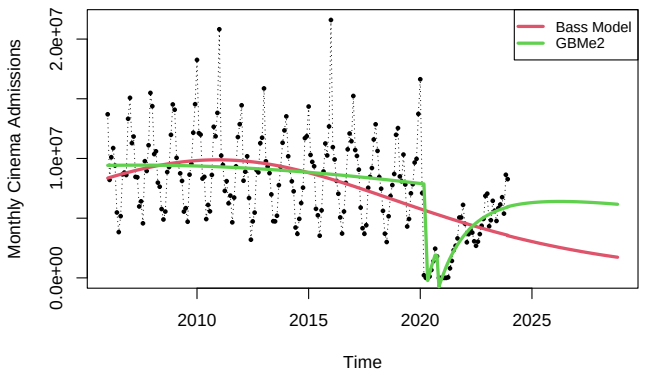




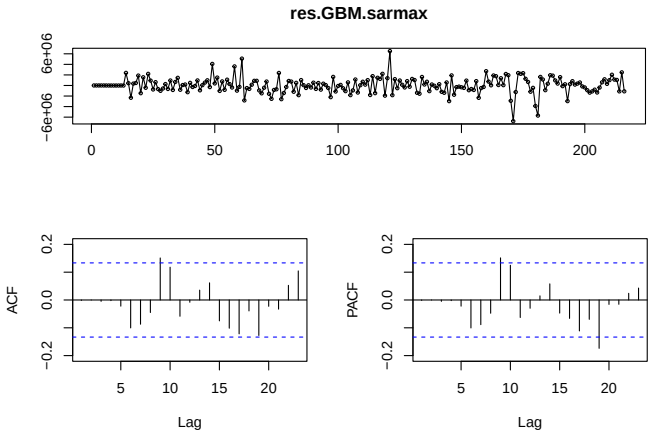


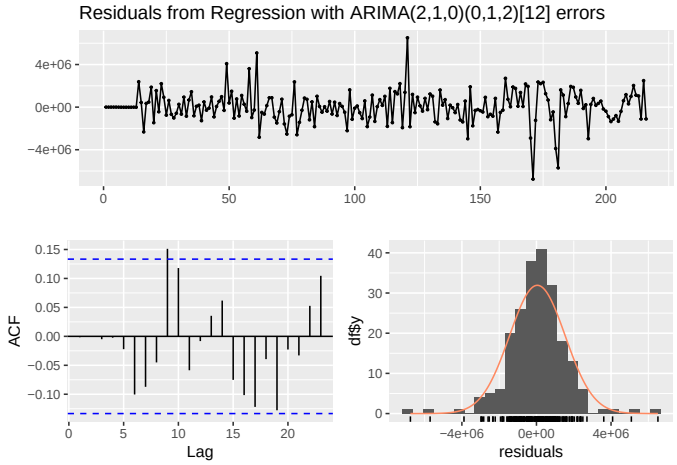


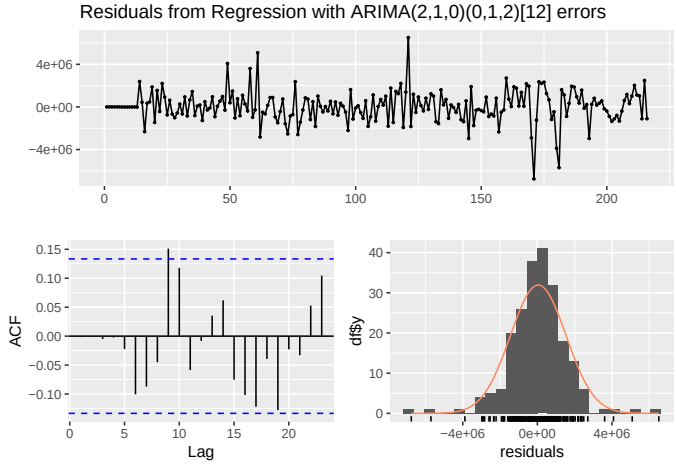
Future Prediction

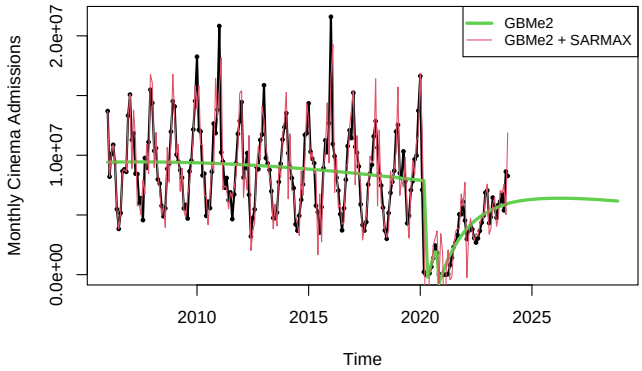


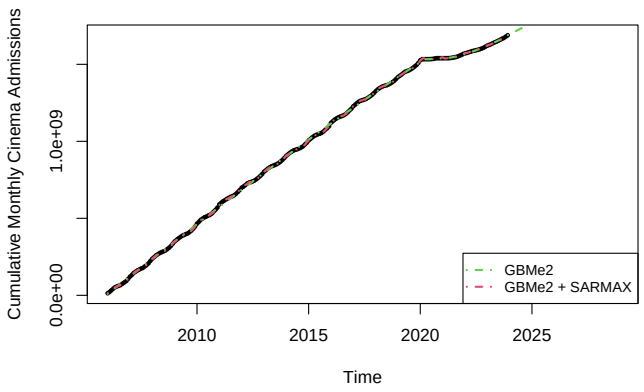
3.3.3 GBM with ARIMAx refinement



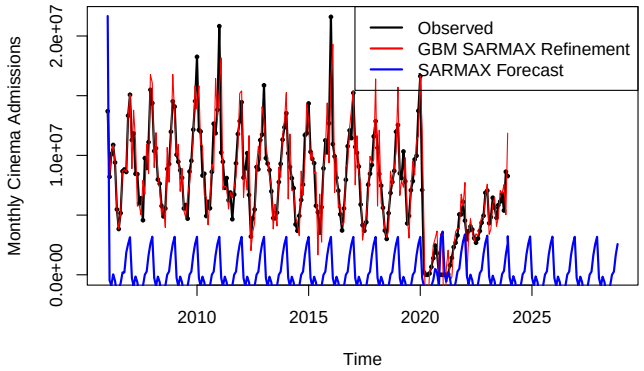






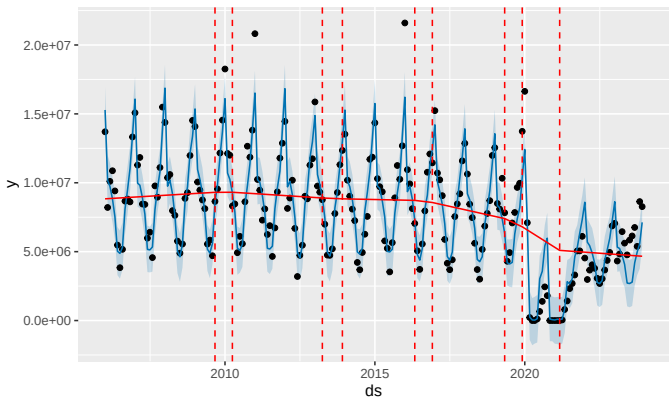


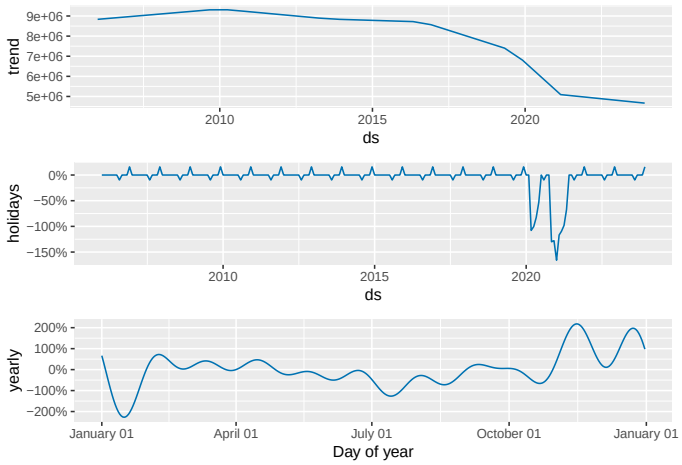
!!! TODO: how to do future predictions when there are xreg???

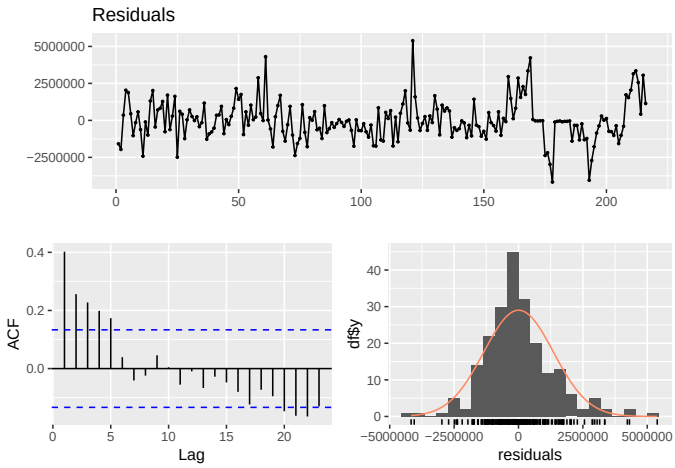


!!! TODO: obviously problematic, i need another solution for the above problem.

3.4 Prophet Model




trend
9e+06
8e+06
7e+06
6e+06
5e+06
2010
2015
2020
ds
holidays
0%
−50%
−100%
−150%
2010
2015
2020
ds
yearly
200%
100%
0%
−100%
−200%
January 01
April 01
July 01
October 01
January 01
Day of year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2500000
0
−2500000
0
50
100
150
200
ACF
0.4
0.2
0.0
0
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10
15
20
Lag
df$y
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30
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10
0
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−2500000
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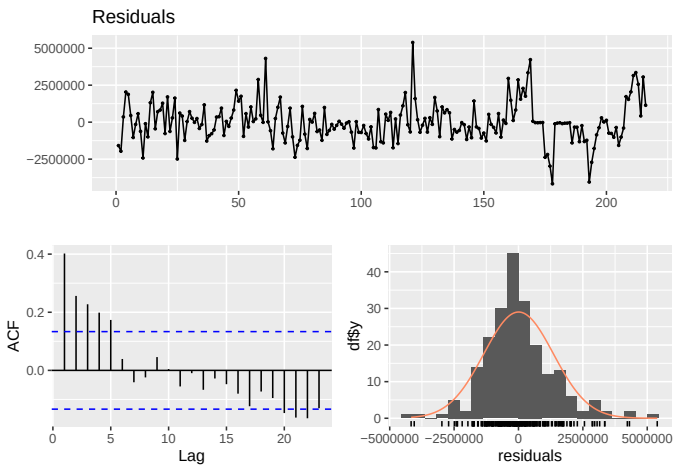
2500000
5000000
residuals


Residuals
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5
10
15
20
Lag
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20
10
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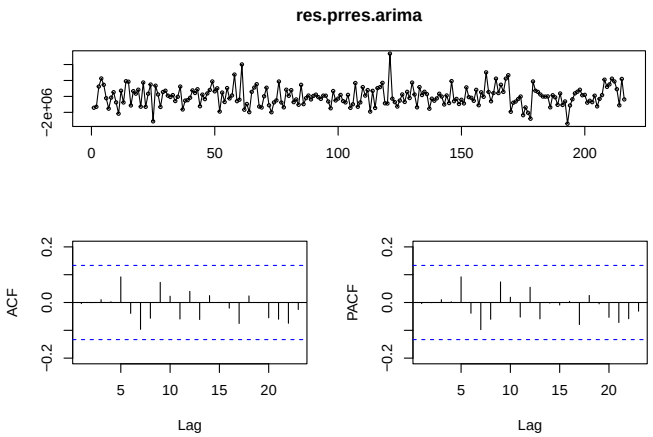
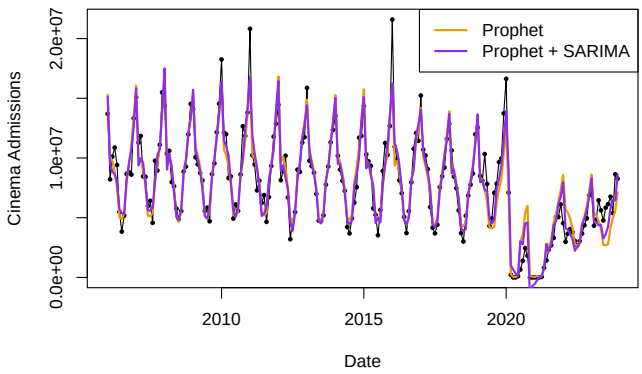
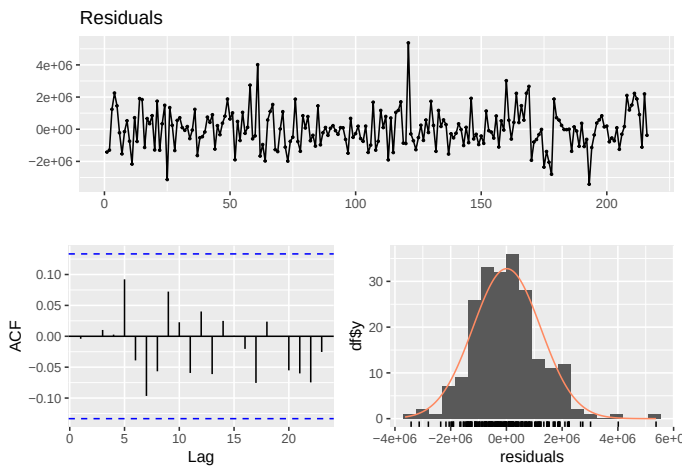
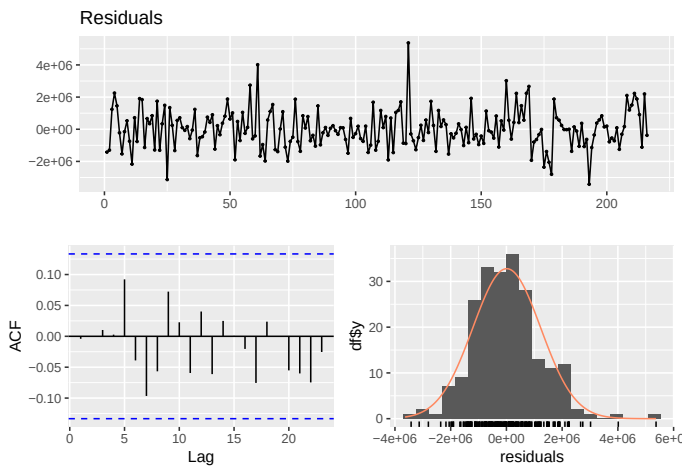
−2500000
0
2500000
5000000
residuals

3.4.1 Prophet with ARIMA refinement



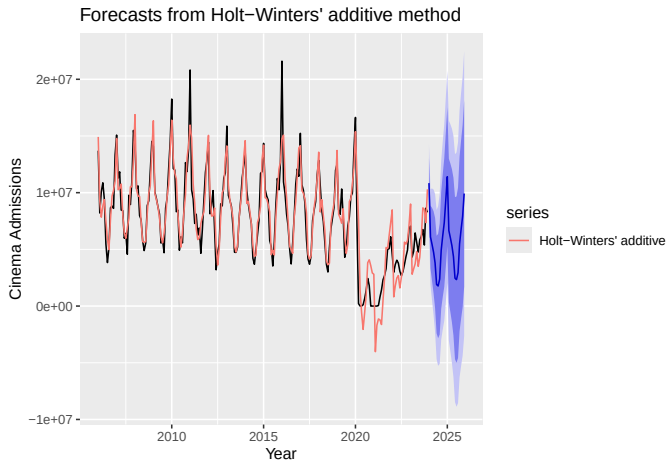


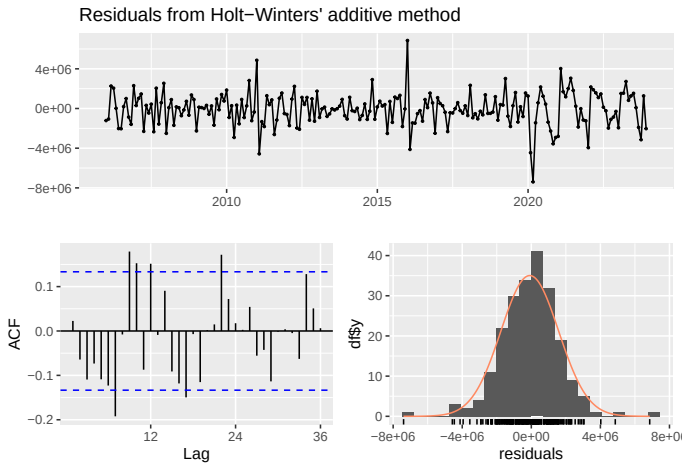


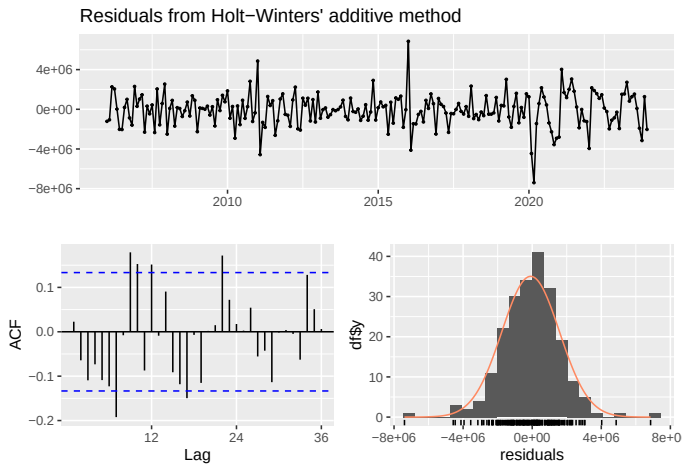


3.5 Exponential Smoothing

3.5.1 Holt-Winters method (for Trend and Seasonality)

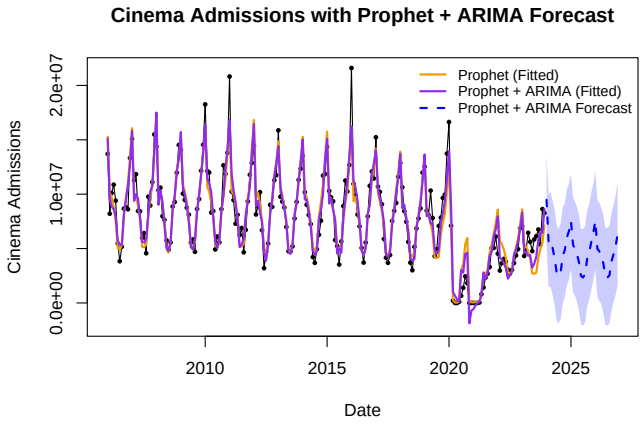






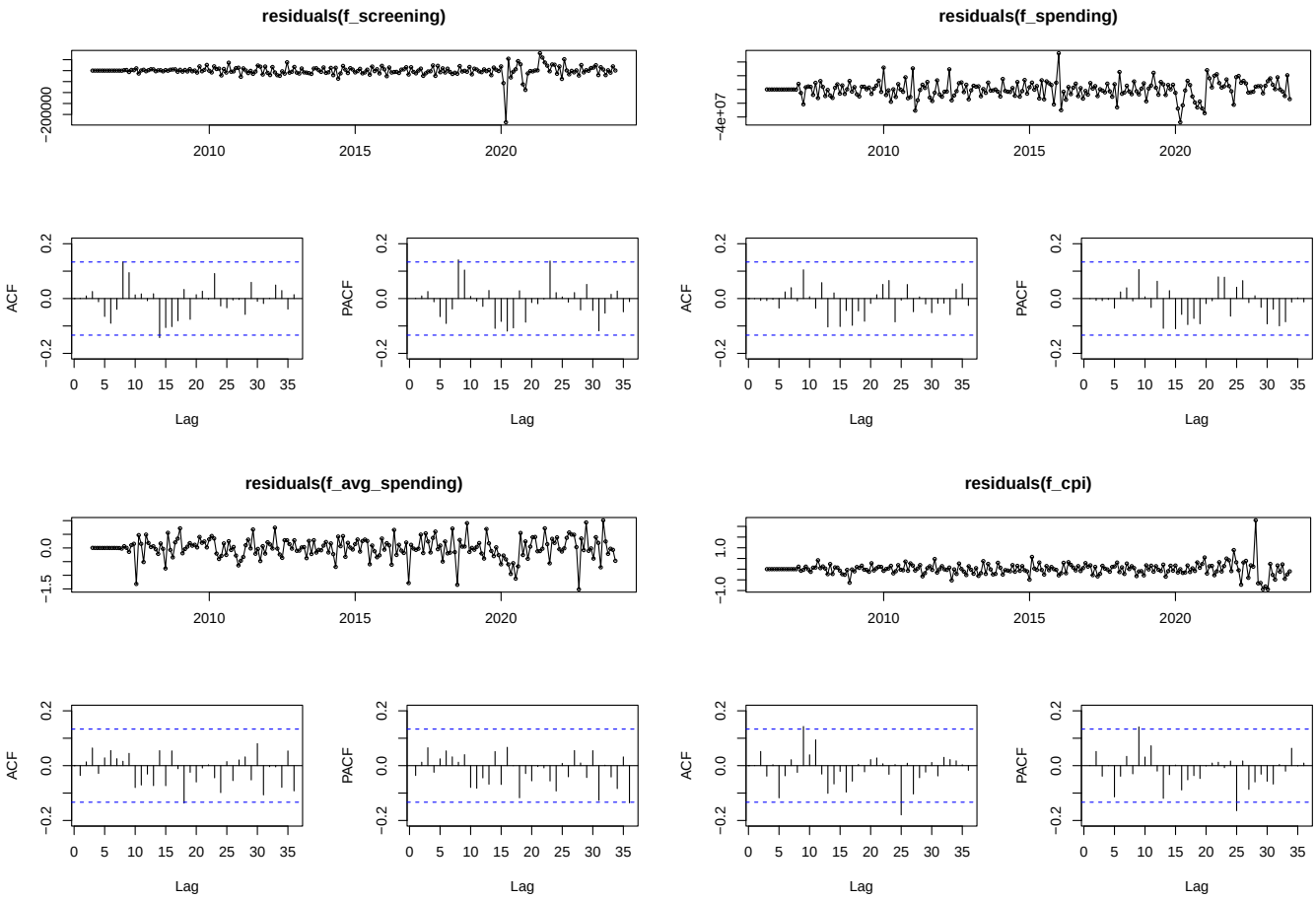
4 3) Forecasting

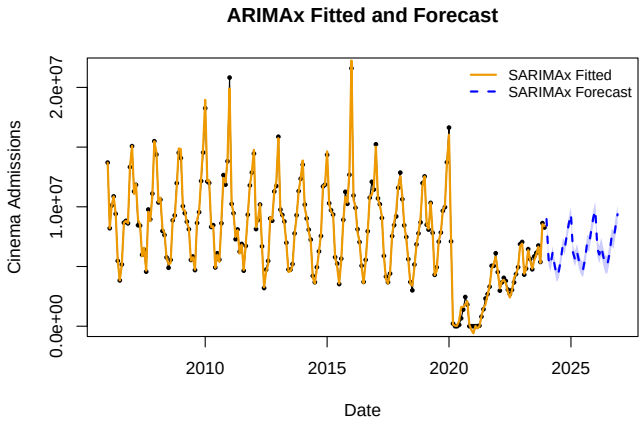
4.1 Forecasting with Prophet + ARIMA



4.2 Forecast with SARIMAx

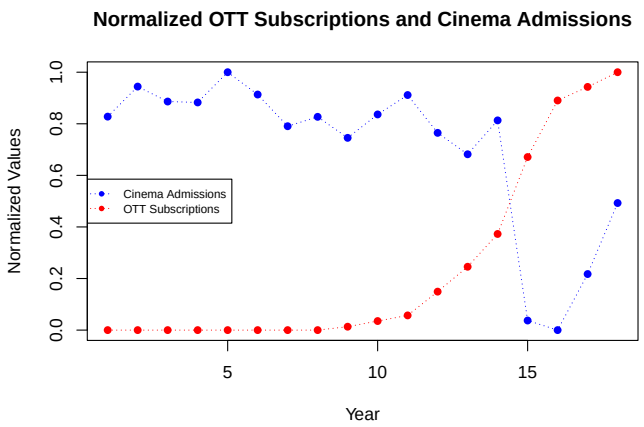
General indication (for differenced data): - If the ACF is exponentially decaying or sinusoidal and there is a significant spike at lag p in PACF and nothing else, it may be an ARIMA(p,d,0). - If the PACF is exponentially decaying or sinusoidal and there is a significant spike at lag q in ACF and nothing else, it may be an ARIMA(0,d,q).



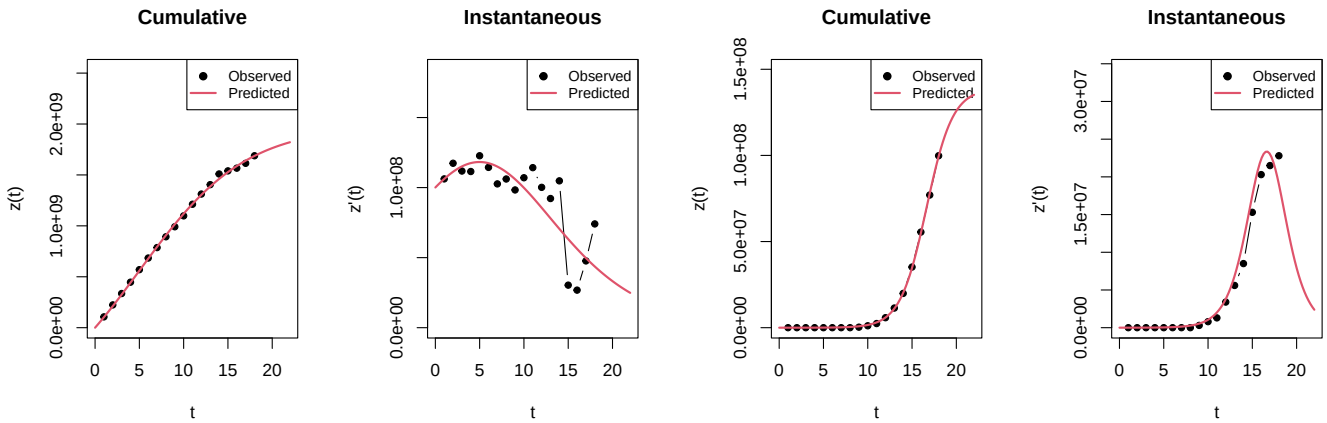


5 4) Unbalanced Competition Regime Change Diachronic model (UCRCD)

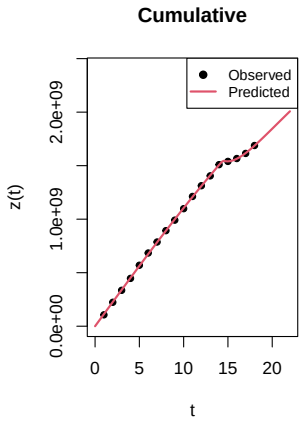
Perform a competition study between OTT and Cinema

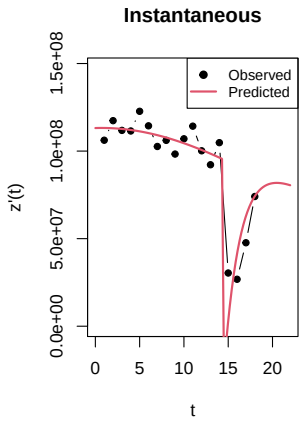


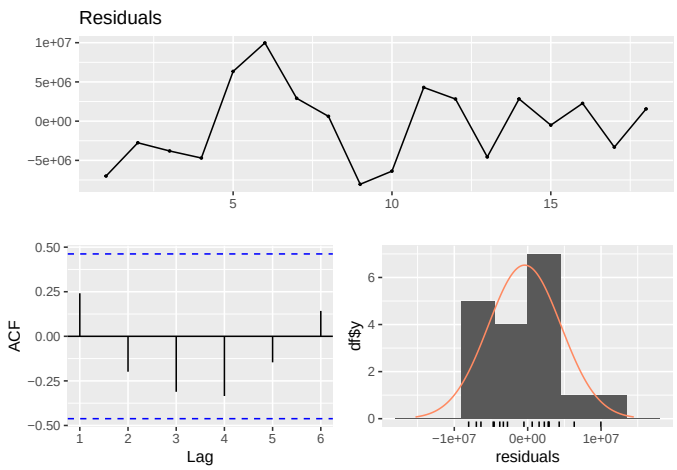
Estimating initial parameters with Bass Model:



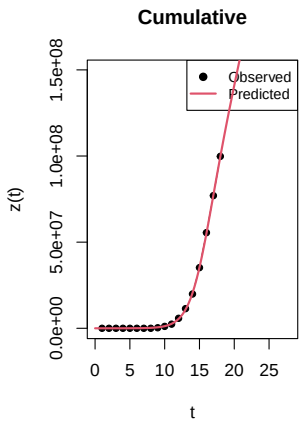
GBM for cinema admissions:

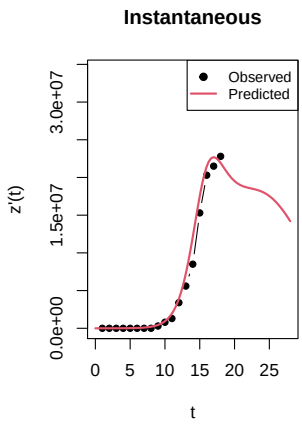


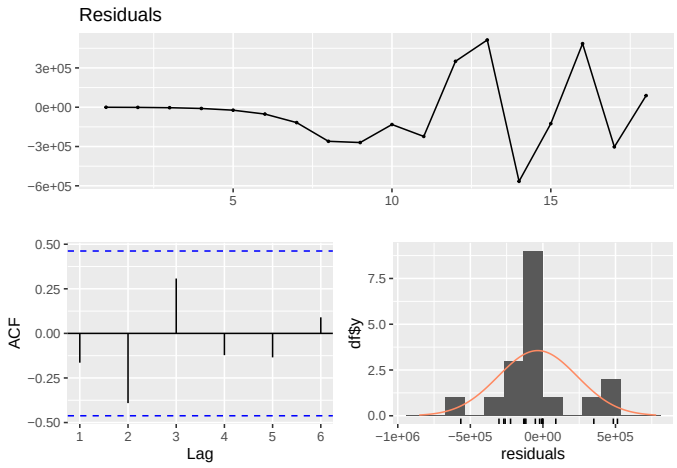


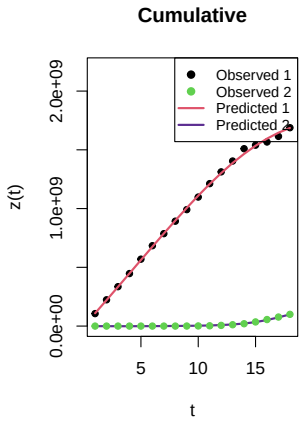


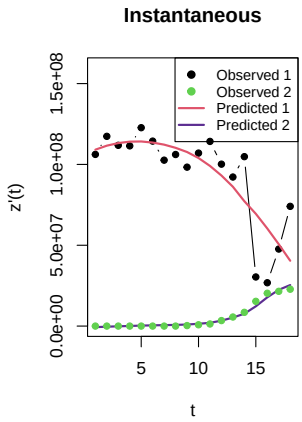
GGM for OTT subscriptions:



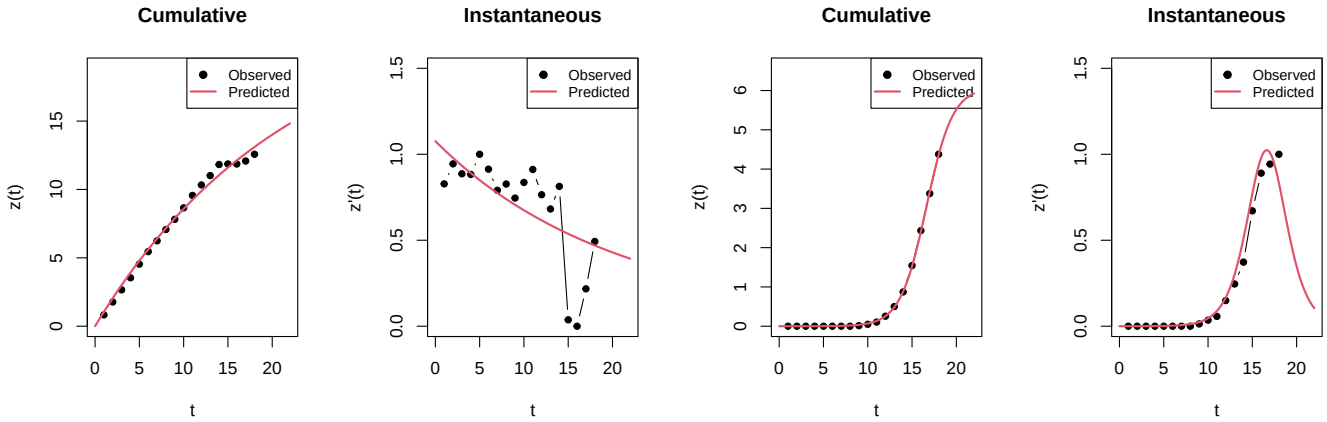




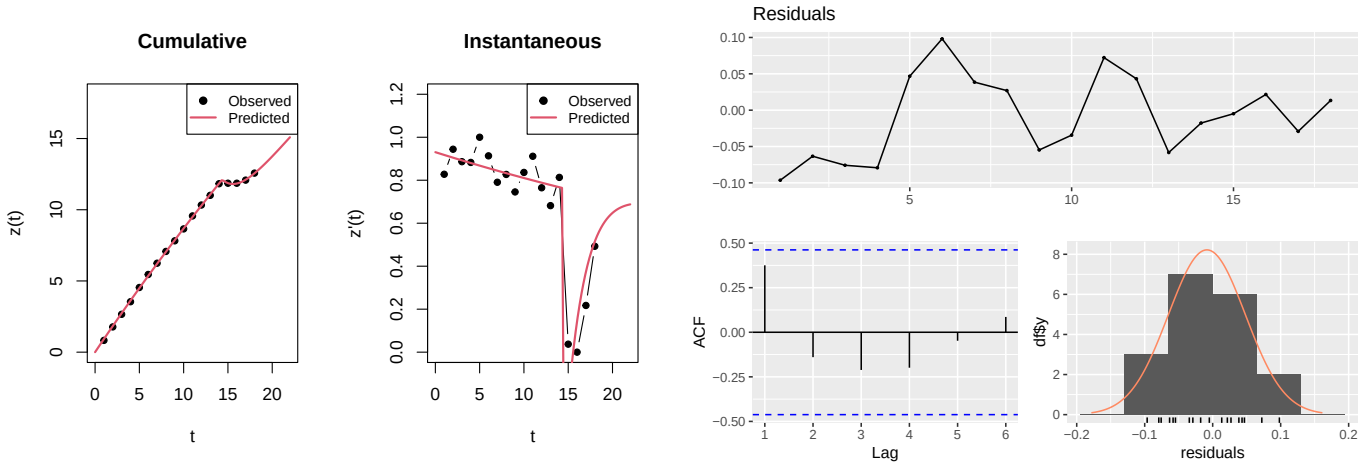




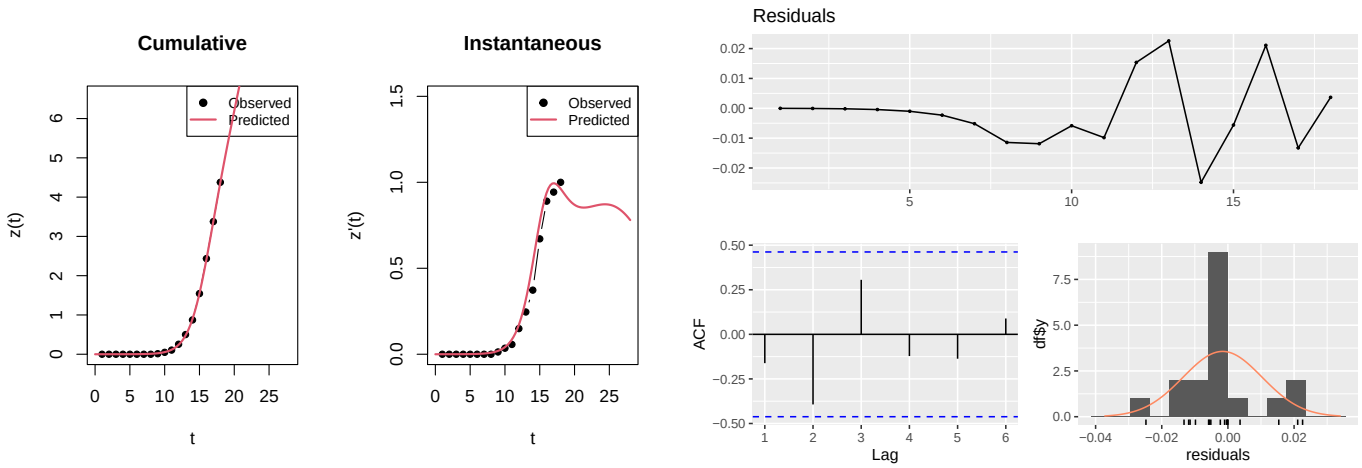
!!! TODO: SINCE THERE IS A SCALE DIFFERENCE IN THE DATA, THE PLOT DOESNT EXPLAIN MUCH. CAN I WORK WITH NORMALIZED DATA WHEN MODELING UCRCD? Will it be academically acceptable? Let's try:

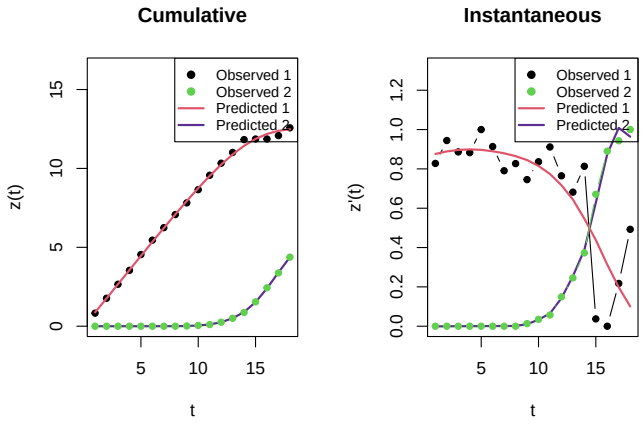


GBM for cinema admissions:



GGM for OTT subscriptions:





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[3] Istituto Nazionale di Statistica (ISTAT). Consumer price index for blue and white-collar worker households. https://esploradati.istat.it/databrowser/.

[4] International Video Federation (IVF). Market information - italy. https://www.ivf-video.org/market-information.

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[7] Google Trends. Google search trends. https://trends.google.com/trends/explore?date=2006-01-01%202023-12-31&geo=IT&q=netflix%20%2B%20amazon%20prime%20video%20%2B%20timvision%20%2B%20disney%20plus.